

Optimal Irrigation for Onion and Celery Production and Spinach Seed Germination in Histosols

Djamila Rekika, Jean Caron,* Guillaume Thérour Rancourt, Jonathan A. Lafond, Silvio J. Gumiere, Sylvie Jenni, and André Gosselin

ABSTRACT

Increasing water scarcity in humid regions requires that food production increase its water use efficiency. Because the hydraulic characteristics of Histosols are different from those of mineral soils, water management for vegetable production must be adapted accordingly. The objective of this research was to determine the optimal soil water potential for irrigating onion (*Allium cepa* L.), celery (*Apium graveolens* L.), and spinach (*Spinacia oleracea* L.) crops in muck soils. Onion and celery were subjected to three irrigation treatments scheduled when tensiometer readings reached -10 or -20 kPa for onion and -30 or -50 kPa (2008) and -15 or -30 kPa (2009) for celery compared with drier control treatments for both crops. For spinach, two irrigation treatments (-10 and -20 kPa) and a control (drier) were tested. Optimal onion marketable yields and jumbo size were obtained from irrigation at potentials above -20 kPa at the bulbing stage. Celery had the best yields with the treatments of 2009 relative to the drier control. The highest spinach germination rate and yield were obtained at -10 kPa. Reliable estimates of the optimal thresholds were consistent with calculations performed using a simple analytical solution to Richards' equation and soil characteristics. Irrigation thresholds for matric potential in a muck soil were shown to be crop specific and could be derived from a model and basic soil hydraulic characteristics.

In the humid regions of southern Quebec, Canada, supplemental irrigation is becoming a more common practice in vegetable crop production. Although the annual rainfall in these regions is about 760 mm, the irregularity of the rainfall distribution is the main reason for using supplemental irrigation to ensure maximum crop productivity and profitability (Gallichand et al., 1991). The proper timing of irrigation is often difficult for growers to determine without useful and objective decision-making tools. Among the various soil-water-based methods, the continuous monitoring of the soil water potential within the root zone is particularly useful for the establishment of adequate irrigation schedules to the extent that it is closely related to plant stress (Hegde, 1986; Marouelli and Silva, 2007; Shock et al., 2000a). Irrigation scheduling by soil water potential criteria was reported to be a powerful method for optimizing plant performance (Shock and Wang, 2011) while minimizing the effects of water stress on the plants

and the leaching of nutrients below the root zone, which can have adverse environmental impacts (Fares and Alva, 2000).

With their shallow root systems, onion, celery, and spinach are sensitive to water stress, and an insufficient water supply during the growing period or seed germination (for spinach) may reduce crop production and quality. Threshold values of soil moisture in mineral soils have been developed for several vegetable crops including onion and celery (Breschini and Hartz, 2002; Phene and Beale, 1976; Phene and Howell, 1984; Shock and Wang, 2011).

In terms of onion response to irrigation water deficits, research has demonstrated that the sensitivity of yield to water stress is dependent on the phenological phase in which water stress is applied (Shock et al., 1998), with the bulb development period being the most sensitive to water stress (Pelter et al., 2004). Shock et al. (2000b) recommended a soil water potential of approximately -20 kPa for an onion crop with drip irrigation in a silty loam soil and showed that total and colossal onion yield increased with decreasing soil water tension to -10 kPa. For furrow-irrigated onion on a silt loam soil, Shock et al. (1998) found that the highest total yield was achieved with -12.5 kPa, but marketable yield was maximized at -27 kPa. Enciso et al. (2009) observed that total yield and marketable jumbo yield were maximized when soil water tension was kept above -30 kPa with drip irrigation in a sandy clay loam soil.

In celery, petiole pithiness (the collapse of parenchyma tissue), a common quality problem, is associated with water stress during

D. Rekika, J. Caron, G. Thérour Rancourt, J.A. Lafond, and S.J. Gumiere, Dep. of Soils and Agri-Food Engineering, Laval Univ., 2480 Hochelaga Blvd., Quebec City, QC, Canada G1V 0A6; S. Jenni, Horticulture Research and Development Centre, Agriculture and Agri-Food Canada, 430 Gouin Blvd., St-Jean-sur-Richelieu, QC, Canada J3B 3E6; and A. Gosselin, Dep. of Phytology, Laval Univ., 2480 Hochelaga Blvd., Quebec City, QC, Canada G1V 0A6. Received 8 May 2013. *Corresponding author (jean.caron@fsaa.ulaval.ca).

Published in Agron. J. 106:981–994 (2014)

doi:10.2134/agronj2013.0235

Copyright © 2014 by the American Society of Agronomy, 5585 Guilford Road, Madison, WI 53711. All rights reserved. No part of this periodical may be reproduced or transmitted in any form or by any means, electronic or mechanical, including photocopying, recording, or any information storage and retrieval system, without permission in writing from the publisher.

Abbreviations: ET, evapotranspiration; ETc, crop evapotranspiration; WUE, water use efficiency.

growth (Aloni and Pressman, 1979). Feigin et al. (1982) reported that yields were highest with optimum soil water potential ≥ -20 kPa in a sandy loam soil, while Kemble and Sanders (2000) recommended a soil water potential of -25 kPa.

For spinach seed germination, high moisture at the surface is essential. Because spinach is shallowly seeded, there may be a rapid fluctuation of soil moisture around the seeds, especially during the warmer months, when an increase in soil temperatures in the seed zone can inhibit seed germination (Leskovar et al., 1999). Germination and emergence of spinach seeds is delayed and reduced as soil water availability decreases, leading to poor stand establishment in the field and consequently to yield losses (Katzman et al., 2001).

In general, irrigation based on the soil water potential depends on climate and soil conditions, irrigation system, crop development stage, and cultivar, with the result that crop- and site-specific thresholds may need to be used (Shock et al., 2007; Shock and Wang, 2011). Attempts should be made to reduce the resources and time required to predict the different thresholds expected in specific systems.

Périard Larrivée (2012) combined calculated evapotranspiration (ET) with soil hydraulic properties to successfully predict the occurrence of a transpiration-related physiological disorder called tip burn in lettuce (*Lactuca sativa* L.) grown on Histosols. Numerous studies have found that crop performance could be related to the proportion of applied water relative to ET on a crop growth cycle basis (Enciso et al., 2007; Gallardo et al., 1996). However, it is expected from ET considerations that plant growth would be optimal if water requirements were regularly met throughout the growing cycle. Outside of rainfall events, this is likely to be the case when a minimal water potential threshold is maintained through irrigation such that the rate of water transfer within the root zone is fast enough to meet spontaneous ET requirements, a condition linked to the unsaturated hydraulic conductivity of the soil and the depth of rooting, among many factors. Early work by Lascano and van Bavel (1984) indicated that transpiration could be related to the bulk soil water potential. Hence, if simple empirical relationships between crop ET and soil hydraulic properties could be derived and if specific irrigation thresholds could be identified, then very significant savings in terms of research efforts could be achieved. This macroscopic analysis may be questioned because the assumption of uniform soil water potential in the near root zone, important to many for the use of such approaches, has also been questioned for many years but neglected because of the difficulties that studying the soil water potential poses (Carminati et al., 2010). Recent observational work has indicated that modification of the water retention and hydraulic properties in rhizosphere soils would result in the absence of a depletion of the water potential near roots up to the root mucilage zone, hence rendering the assumption of uniformity within the bulk soil reasonable (Carminati et al., 2010). Consequently, a relationship between bulk soil water potential and transpiration may be successfully determined (Lascano and van Bavel, 1984; Périard Larrivée, 2012) and such macroscopic model assumptions met.

Despite the large number of studies, most have dealt with crops grown on mineral soils, while few studies have been performed to determine the optimum soil water potential

or to evaluate hydrologic performance and related irrigation requirements on muck soils. This study dealt with the optimization of irrigation management based on soil water potentials for onion, celery, and germinating spinach grown in muck soils. Specifically, the objectives were to determine the yield and quality response of these crops to soil water potential irrigation thresholds and to establish links between soil hydraulic properties and the identified thresholds. Numerical models could be successfully used to derive such thresholds (Périard et al., 2013), but simple steady-state analytical solutions (Yuan and Lu, 2005) requiring few parameters could be easily implemented to achieve such goal. The comparative performances of such analytical solutions have never been evaluated. Such an approach may be relevant to any scientist or grower who is interested in irrigation of shallow root system crops grown on muck soils or peat growing media and possibly extended to other soil types.

MATERIALS AND METHODS

Experimental Site and Crop Details

The experiments were performed in commercial fields at the Coop des Producteurs Multiveg farms in southwestern Quebec, Canada ($45^{\circ}12' \text{ N}$, $73^{\circ}27' \text{ W}$) on muck soils (Histosols). These Histosols, classified mainly as Haplohemists (Soil Survey Staff, 1999) or as Mesisols in the Canadian system of soil classification (Soil Classification Working Group, 1998), were put into agricultural use about 50 yr ago by digging ditches and installing drain tiles at a depth of about 1 m. These Histosols are usually stratified, with a more decomposed layer (humic horizon) from the surface to around the 25-cm depth, often followed by a 10- to 20-cm-thick humic compacted layer over a fibric or mesic layer deeper than 35 to 45 cm. Under the muck soil profile, sometimes deeper than 3 m, an impermeable layer made of clay to mixed sand and clay is found in these Quebec Histosols. The studied crops were short-day yellow onion (cultivars Norstar in 2008 and Timberline in 2009), celery (cultivar XP266), and spinach (cultivars Waitiki, C2-606, and Toucon). Nitrogen fertilizer was applied at a rate of 130 kg ha^{-1} for onion and 110 kg ha^{-1} for celery, with 50 and 75% preplant incorporated and 50 and 25% sidedressed in accordance with the current recommendations (Conseil des Productions Végétales du Québec, 1996). Potassium and P were applied according to soil tests and recommendations.

Onion cultivars were seeded on 1.83-m beds, each consisting of four double rows spaced 0.25 m apart, on 21 Apr. 2008 and 30 Apr. 2009. The growing periods covered 125 d in 2008 (until 27 August) and 124 d in 2009 (until 31 August). Each double row consisted of two onion rows spaced 0.15 m apart and seeded at a rate of seven seeds per 0.3 m. The plots were six beds wide and 9.1 m long.

Celery plants were transplanted in the field on 10 July 2008 and 13 July 2009, with an in-row spacing of 0.25 m and a between-row spacing of 0.61 m to give an approximate density of 5 plants m^{-2} . The plots measured 9.1 by 5.5 m. The celery plants were immediately irrigated after planting to ensure their survival. The growing periods covered 85 d in 2008 (until 3 October) and 79 d in 2009 (until 30 September).

Baby spinach was sown from 15 June to 9 Aug. 2010 on 1.75-m-wide beds, with 24 rows spaced 5 cm apart. The seeding rate was 49 seeds per linear meter. The experimental plot was

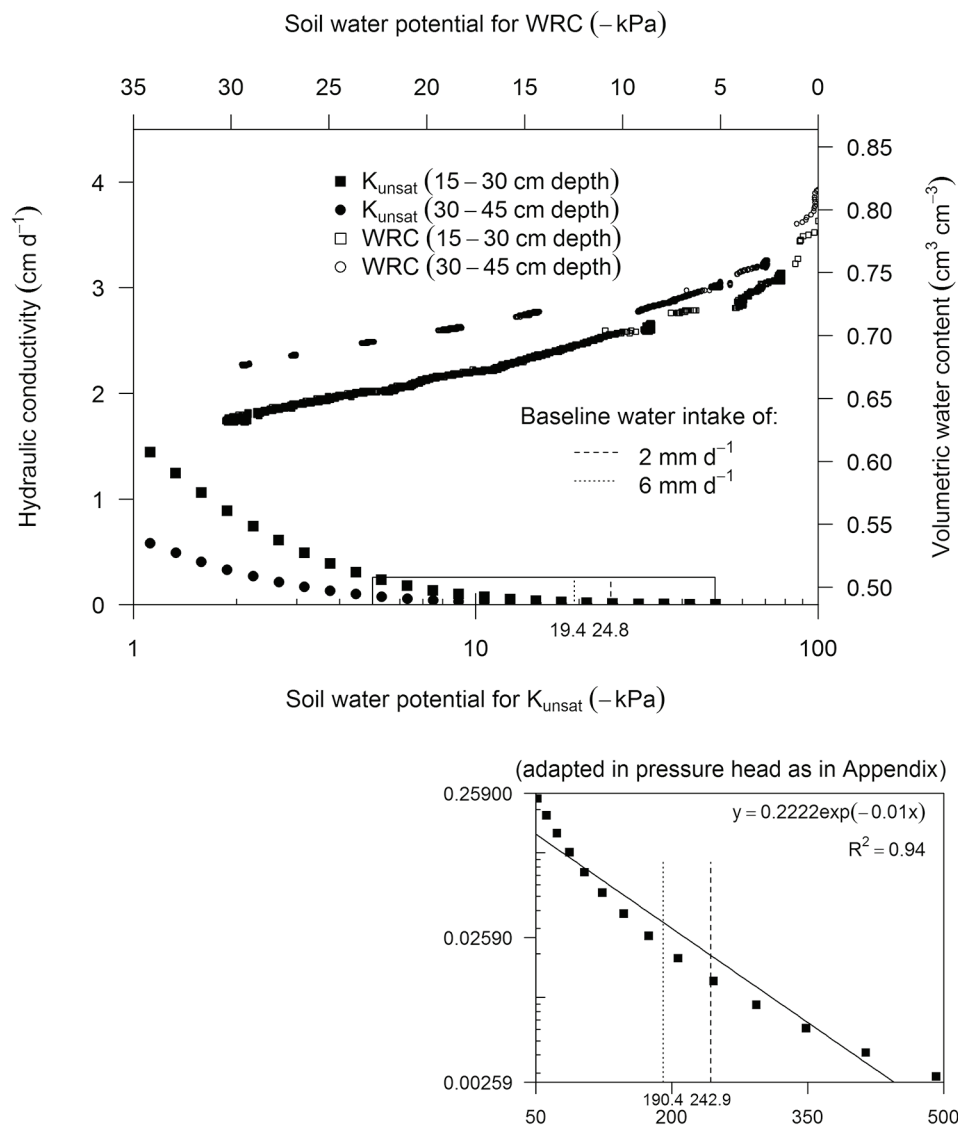


Fig. 1. Unsaturated hydraulic conductivity (K_{unsat}) and volumetric water content at different matric potentials for yellow onion (cultivars Norstar in 2008 and Timberline in 2009). Water retention curves (WRC) represent the multistep outflow data on top and right axes, while K_{unsat} curves are on left and bottom axes; thresholds are shown in the insert graph in the same units as in the appendix.

three raised beds wide by 18.1 m long. All fields were immediately irrigated after sowing to stimulate seed germination.

Irrigation Treatments and Experimental Design

The irrigated and unirrigated onion plots were arranged in a randomized complete block design with four blocks. The irrigation treatments consisted of irrigation throughout the growing season, irrigation during the bulbing stage, and a control (unirrigated, with natural rainfall only). Soil water potentials were measured using four wireless electronic tensiometers (TX-80-WL and TX3, Hortau Inc.) inserted at 0.2- and 0.3-m depths below one onion row in each irrigation treatment and recording continuously at 15-min intervals on a control system (Irrolis-WEB, Hortau Inc.) used for making decisions on irrigation. The irrigation treatments were applied whenever two or three consecutive measurements indicated that the average soil water potential readings at the 0.2-m depth had reached the set point of -20 kPa for each treatment.

The celery experiment was set up in a randomized complete block design with two and three blocks in 2008 and 2009,

respectively, and three irrigation treatments: two irrigation thresholds (-30 and -50 kPa) and a control (irrigation based on the growers' personal experience) in 2008, and two irrigation thresholds (-15 and -30 kPa) and a control (no irrigation) in 2009. The effective depth of watering was 0.15 m during the first 25 d after transplanting, 0.25 m during the following 24 d, and 0.35 m thereafter until harvest. Irrigation was started when the average of all the soil water potential readings from all plots in a treatment reached the expected threshold. Applied water was measured by four rain gauges per irrigation treatment.

In the spinach experiment, two irrigation thresholds and a control (irrigation based on the growers' personal experience) were evaluated and distributed randomly in the field. Irrigations were initiated on the basis of soil water potentials monitored in the first 0.03 m of the soil layer when thresholds of -10 and -20 kPa were reached.

All crops were irrigated using an overhead sprinkler system.

Crop evapotranspiration (ET_c) was calculated a posteriori using the FAO Penman–Monteith equation (Allen et al., 1998, p. 90) by multiplying reference evapotranspiration

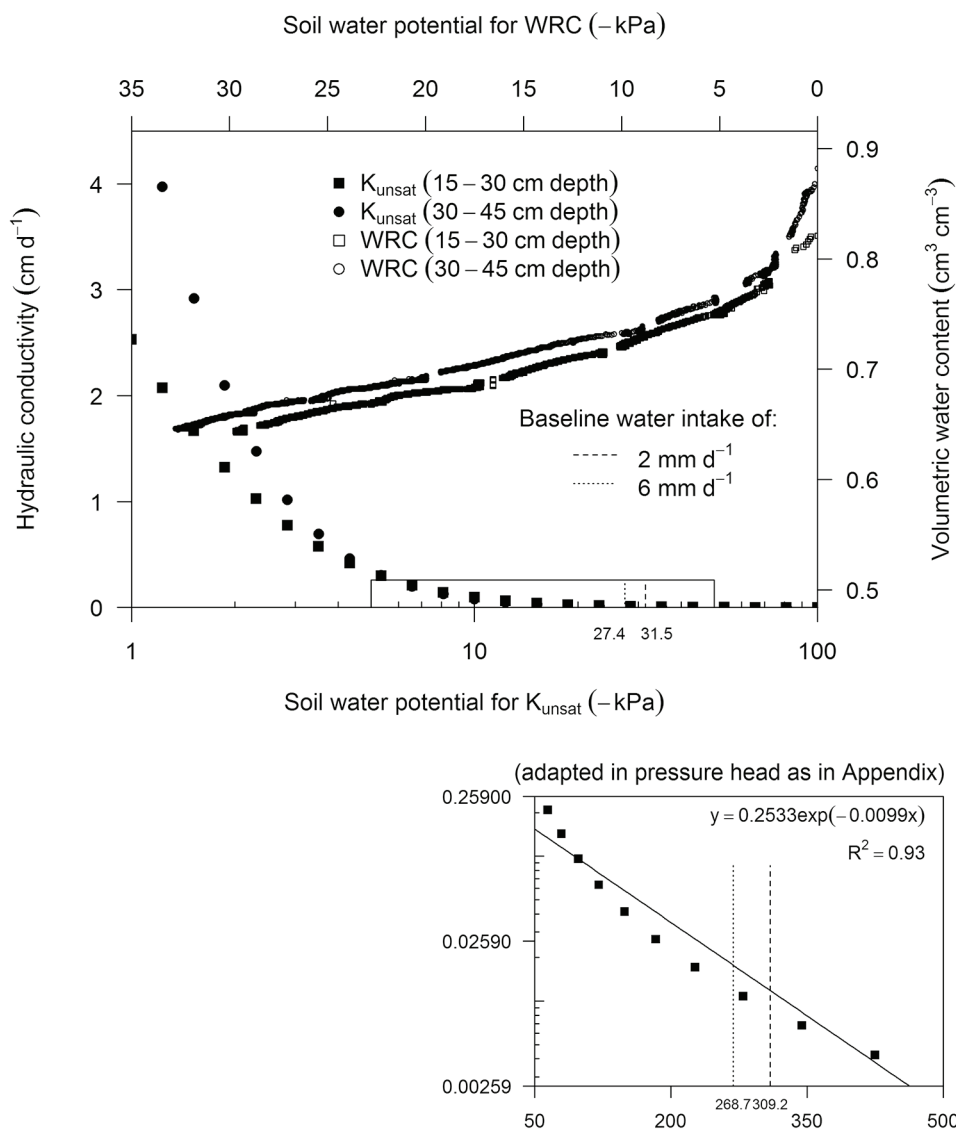


Fig. 2. Unsaturation hydraulic conductivity (K_{unsat}) and volumetric water content at different matric potentials for celery (cultivar XP266). Water retention curves (WRC) represent the multistep outflow data on top and right axes, while K_{unsat} curves are on left and bottom axes; thresholds are shown in the insert graph in the same units as in the appendix.

(ET_o) by the corresponding crop coefficients (K_c). Reference evapotranspiration was calculated from weather data collected within 3 km of the sites (Allen et al., 1998, p. 24); K_c values were obtained from Allen et al. (1998, Table 12, p. 110).

Crop water use efficiency (WUE) was calculated as the ratio of marketable yield (Mg ha^{-1}) to the amount of water received including irrigation and rainfall (mm).

Plant Measurements

When the onions reached maturity, four 1- to 2-m-long sections from a middle row of each replicate were harvested by hand in 2008 and 2009. The onions were left in windrows to dry for 7 to 14 d, depending on the weather each year. After drying, plant tops were manually removed at a point approximately 3 cm above the bulb using scissors. Rotten and split bulbs were separated before grading. The bulbs were classified by size as prepack (<45 mm), small (45–57 mm), medium (57–76 mm), or jumbo (>76 mm). For each size class, the weight was recorded and the yield was computed.

Marketable onions were small, medium, and jumbo size classes. Total yield included prepack, rotten, and split bulbs.

Celery was harvested on 1.75-m-long sections of two rows in each plot. After recording the untrimmed weight and plant height, the celery was trimmed to a uniform length of 37 cm, and unmarketable petioles were removed to provide fresh celery yields. Each section was reweighed (trimmed weight), plant diameters were measured, and marketable petioles were counted. In accordance with the criteria of the Canadian Food Inspection Agency (1999), marketable celery plants (Canada no. 1 grade) had a minimum stalk length of 30.5 cm and a minimum diameter of 6.4 cm. Final harvest occurred on 3 Oct. 2008 and 30 Sept. 2009.

Emerged spinach seedlings were counted 7 d after planting, i.e., when they had reached the cotyledon stage. Seedlings were counted using three 0.5-m² quadrats randomly distributed in each plot and each irrigation treatment. The baby spinach was mechanically harvested, and yield was determined from 22.8-m² sections (24 rows, each 13 m long).

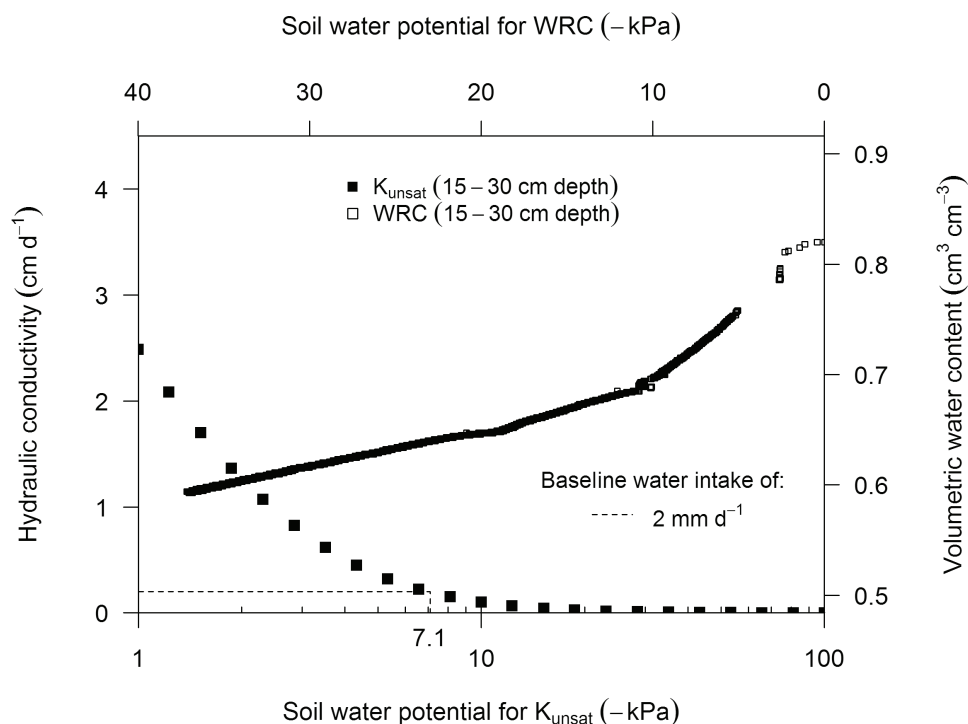


Fig. 3. Unsaturated hydraulic conductivity (K_{unsat}) and volumetric water content at different matric potentials for spinach (cultivars Waitiki, C2–606, and Toucon). Water retention curves (WRC) represent the multistep outflow data on top and right axes, while K_{unsat} curves are on left and bottom axes.

Water Desorption Curve, Unsaturated Hydraulic Conductivity, and Threshold Estimation

To determine the influence of physical soil properties on the water supply of irrigated crops, unsaturated hydraulic conductivity and soil water desorption curves were determined at two soil depths. Despite the fact that all experiments were performed in the same region and soil type, the soil physical characteristics were different among the fields. Samples were saturated from bottom to top for 24 h, then the saturated hydraulic conductivity was measured using a vertical constant-head soil core method (Reynolds and Elrick, 2002) slightly modified to enable automatic measurements. Subsequently, Tempe pressure cells (Soilmoisture Equipment Corp.) equipped with tensiometers were used to determine the soil water desorption curves (Dane and Hopmans, 2002). Samples were weighed at saturation and at the end of the multistep outflow test, then dried at 70°C for 24 to 48 h. Total porosity was assumed from saturated water content measurements and allowed the calculation of the soil volumetric water content from the outflow volumes. This multistep outflow method (Hopmans et al., 2002) also allowed the calculation of unsaturated hydraulic conductivities by inverse modeling in HYDRUS-1D (PC-Progress). Only the measured water retention curves and the modeled unsaturated conductivity curves related to the specific fields used for onion (Fig. 1) and celery (Fig. 2) in 2009 and Field 4 used for spinach seed germination (Fig. 3) are presented here.

For deriving threshold estimates in the onion and celery experiments, the matric potential form of Richards' equation (Elrick and Reynolds, 1992) and a Gardner function (Gardner, 1958; adapted in pressure head units [cm]; see appendix) were used to approximate the unsaturated hydraulic conductivities for specific ranges obtained in HYDRUS-1D by the Mualem

hydraulic conductivity function because it corresponded to the observed data in the matric potential (ψ) domain observed before irrigation initiation (–5 to –50 kPa range, see below). Solutions under steady-state and transient-state conditions, taking into account root uptake and rooting depth, have been published (Yuan and Lu, 2005) and may allow a reasonable approximation of critical irrigation thresholds. Moreover, such an approach may also provide a solution for a situation where the unsaturated hydraulic conductivity is piecewise exponential through the whole matric potential domain (Caron et al., 2005).

Evapotranspiration values of 2 mm d^{–1} (evaporation dominated) and 6 mm d^{–1} (maximum ET observed in the field) were used for deriving the critical irrigation threshold for a given rooting depth with corresponding unsaturated hydraulic conductivity value (see appendix). The implementation of the derivation of such a threshold from a set of unsaturated hydraulic conductivity and pressure head data stored in an Excel file was performed using Maple software (Maple 16.00, Maplesoft). The program was used to select the appropriate linear range and implement it in critical pressure head (h_c) calculations (see appendix).

Estimation of soil water flux within the root zone of a crop is necessary for quantifying plant water uptake use, especially during dry spells, in an attempt to reduce the risk of excessive irrigation and nutrient leaching to the groundwater. Soil water flux was calculated according to the Buckingham–Darcy equation:

$$q_z = -K(h) \frac{d(h+z)}{dz} \quad [1]$$

where q_z is the water flux (cm d^{–1}) for 1-h intervals, h is the pressure head (–cm) obtained after conversion of the observed matric potential (–kPa), $K(h)$ is the unsaturated hydraulic

Table 1. Irrigation, rainfall, soil water potential, flow of water in the soil, seasonal crop evapotranspiration (ETc), and water use efficiency (WUE) for yellow onion (cultivars Norstar in 2008 and Timberline in 2009) and celery (cultivar XP266).

Irrigation treatment	Avg. soil water potential kPa	Avg. water flux			Seasonal ETc mm (mm d ⁻¹)	Irrigation mm	Rainfall mm	WUE† Mg ha ⁻¹ mm ⁻¹
		In the soil	During leaching	During capillary rise				
			cm d ⁻¹					
Onion, 2008								
Growing season	-8.7 a‡	NA§	NA	NA	379 (3.0)¶	75	420	0.16
Bulbing	-8.3 a	NA	NA	NA		75		0.16
Control	-18.1 b	NA	NA	NA		0		0.18
Onion, 2009								
Bulbing	-8.3 a	-0.24	-0.51	0.15	375 (3.0)	60	352	0.17
Control	-22.0 b	7.8 × 10 ⁻⁴	-0.17	0.08		0		0.18
Celery, 2008#								
-30 kPa	-14.2 a	5.1 × 10 ⁻⁴	-2.5 × 10 ⁻⁵	5.7 × 10 ⁻⁴	196 (2.3)	55	246	0.19
-50 kPa	-21.4 b	2.7 × 10 ⁻⁵	-4.4 × 10 ⁻⁴	2.5 × 10 ⁻⁴		25#		0.25
Control	-26.3 c	1.1 × 10 ⁻³	0.00	1.1 × 10 ⁻³		55		0.21
Celery, 2009								
-15 kPa	-11.0 a	0.22	-0.62	0.73	223 (2.8)	125	184	0.21 b
-30 kPa	-20.4 b	-0.04	-0.76	0.41		112		0.21 b
Control††	-66.1 c	0.15	-0.70	0.49		25#		0.26 a

† Calculated as marketable yield divided by the amount of water received (irrigation plus rainfall).

‡ For each year, means within each column followed by the same lowercase letter are not significantly different using least significant difference in PROC MIXED of SAS at $p \leq 0.05$. In columns that have no lowercase letters following the means, the treatments were nonsignificant.

§ NA, not available.

¶ Average daily ETc in parentheses.

Matric potential reading average from 22 Aug. to 18 Sept.

†† Water applied for establishment of the crop.

Table 2. Effect of two irrigation levels on yellow onion (cultivars Norstar in 2008 and Timberline in 2009) yield parameters as classified by size during the two growing seasons of 2008 and 2009.

Irrigation treatment	Total yield	Prepack (<45 mm)	Small ($45\text{--}57$ mm)	Medium ($57\text{--}76$ mm)	Jumbo (>76 mm)	Irrigations
2008						
Growing season	77.8 ab†	0.2b	7.2	63.4	5.3 a	4
Bulbing	78.8 a	0.1b	5.8	65.3	6.3 a	4
Control	73.5 b	0.5a	8.0	61.1	2.1 b	0
2009						
Bulbing	71.0 a	0.0	0.4	61.5	7.1 a	3
Control	65.1b	0.2	0.9	60.0	1.9 b	0

† Within each year, means followed by the same lowercase letter across columns are not significantly different using least significant difference in PROC MIXED of SAS at $p \leq 0.05$. Where no lowercase letters follow the means, no significant differences were found.

Table 3. Plant parameters and marketable yields of celery (cultivar XP266) trimmed to a 37-cm height under three irrigation regimes during the two growing seasons of 2008 and 2009.

Irrigation treatment	Weight per plant		Untrimmed plant height	Stalk diameter	Marketable petioles	Marketable yield	Irrigations
	Untrimmed	Trimmed					
	g		cm		no.	Mg ha ⁻¹	no.
2008							
−30 kPa	NA†	903	77.0	09.9	15.4	56.7	2
−50 kPa	NA	1012	78.1	10.4	16.0	66.4	0
Control‡	NA	973	78.5	10.3	16.4	63.6	2
2009							
−15 kPa	1820 a§	969 a	84.7 a	12.9 a	16.3	63.5 a	6
−30 kPa	1718 a	954 a	80.8 b	12.5 a	15.0	62.6 a	5
Control¶	1265 b	816 b	71.3 c	11.2 b	15.5	53.5 b	0

† NA, not available.

‡ Grower's irrigation.

§ Within each year, means followed by the same lowercase letter across columns are not significantly different using least significant difference in PROC MIXED of SAS at $p \leq 0.05$. Where no lowercase letters follow the means, no significant differences were found.

¶ No irrigation.

conductivity (cm d^{-1}) specific to each field, $d(h+z)/dz$ is the hydraulic gradient (dimensionless), and z is the position (cm) in the soil profile. The $d(h+z)$ term in Eq. [1] was calculated as the difference, for the considered time interval, in matric head between the two tensiometers monitoring the suction below the surface and at a given depth within the profile, where dz is the difference in depth between the two sensors. The first and last value of the tensiometer data were averaged at each separate depth z before calculating the gradient. A negative q_z value indicates a downward water flux (leaching), whereas a positive value indicates an upward flux (capillary rise). To find the critical matric potential for a capillary rise of 2 mm d^{-1} required for a given seed planted 3 cm below the surface, we assumed a fixed (unitary) value for the hydraulic gradient. Close to the surface, an assumed value was preferred over a measured gradient because measurements would have required placing a top tensiometer close to the surface, hence having the risk of a potential measurement influenced by surface evaporation, characteristic of a dry surface. Such a surface measurement would bear a high risk of biasing the gradient. Therefore, it was reasonable to assume a unitary gradient for liquid flow because following seeding, evaporation would dominate the flux (with liquid flow below the surface up to a point where vapor flow takes over) and root water uptake would be equal to zero. Finally, it was reasonable to assume liquid flow to the seed because the seeds were buried 3 cm below the surface and liquid flow dominates within the -5 to -30 kPa range observed at that depth.

Statistical Analysis

Data were statistically analyzed using the MIXED procedure of the SAS software package (SAS Institute, 2008). The least significant difference test ($p = 0.05$) was used for mean separation, and the letters indicating significant differences were assigned using the macro PDMIX 800 procedure (Saxton, 1998). Stalk diameter, untrimmed plant height, marketable petioles, and weight per plant for celery and jumbo and medium grades for onion were logarithmically transformed to stabilize variances and to normalize the data sets. Spinach seed germination data were statistically analyzed using the GLM procedure of SAS. Treatment means were compared using the least square means (LSMEANS) procedure of SAS.

RESULTS AND DISCUSSION

Management of Irrigation

Rainfall during the growing season (approximately 20 April–5 October) was 490 mm in 2008 and 457 mm in 2009, less than the 30-yr normal (562 mm) during this period. The ET_c during the growing periods were determined to be 379 and 375 mm in 2008 and 2009, respectively, for onion and 223 mm in 2009 for celery (Table 1).

In both years, adequate rainfall during the vegetative growth of onion should have minimized the need for irrigation. On some occasions, however, stress periods occurred and overlapped the bulbing and enlargement stages of onion as reflected by soil matric potential increases (Fig. 4), and irrigation was needed four and three times for total amounts of water received of 75 and 60 mm in 2008 and 2009, respectively (Tables 1 and 2). The season average of soil water potentials at the 0.2-m depth for onion in the irrigated plots (bulbing and

enlargement stages) was around -8 kPa , significantly higher than in the unirrigated plots, for which values were -18 and -22 kPa depending on the year (Fig. 4; Table 1).

For celery, the total amount of water supplied by irrigation in 2008 was 30 mm for both the control and the -30 kPa treatments, resulting in no significant differences between treatments. Such results were consistent with abundant rainfall, distributed well enough so that no irrigation occurred for the -50 kPa treatment because this threshold was not reached. Unlike in 2008, the thresholds of -15 and -30 kPa were reached several times in 2009 (Fig. 5), resulting in six and five irrigations (Table 3) and total water amounts of 100 and 87 mm, respectively (Table 1). In contrast, the season average of soil water potentials in the unirrigated plots remained low (-66 kPa) during the dry period relative to the irrigated plots (-11 to -20 kPa), probably as a result of total rainfall lower than the crop ET . Also, there were several days when the soil water potential was at -80 kPa or more (Fig. 5), reflecting a very likely water stress experienced by the plants when water tended to be depleted near the root zones, as reported by Périard Larrivière (2012).

Table 4. Effect of irrigation treatments on seed germination and yield of spinach (cultivars Waitiki, C2–606, and Toucan all pooled together) grown in 2010.

Field	Irrigation treatment	Seed germination plants m^{-2}	Yield g m^{-2}	Irrigations no.	Total water applied mm
1†	-10 kPa	731	NA‡	0	0
	-20 kPa	748	NA	0	0
	control	709	NA	0	0
2	-10 kPa	650 a§	869	2	8
	-20 kPa	624 b	799	0	0
	control	621 b	823	0	0
3†	-10 kPa	709	NA	0	0
	-20 kPa	697	NA	0	0
	control	700	NA	0	0
4	-10 kPa	417	753	3	12
	-20 kPa	379	667	2	8
	control	367	699	1	4
5†	-10 kPa	524	NA	0	0
	-20 kPa	552	NA	0	0
	control	534	NA	0	0
6	-10 kPa	798 a	NA	4	16
	-20 kPa	692 b	NA	1	4
	control	710 b	NA	1	4
7	-10 kPa	691	NA	2	8
	-20 kPa	688	NA	1	4
	control	716	NA	1	4
8	-10 kPa	751	NA	2	8
	-20 kPa	703	NA	0	0
	control	713	NA	1	4
Avg.	-10 kPa	661 a	811 a		
	-20 kPa	617 b	733 c		
	control	626 b	761 b		
<i>p</i> value		0.0330	0.0097		

† Due to rainfall, thresholds were not reached in this field and no irrigation was performed.

‡ NA, not available.

§ Data followed by the same lowercase letter are not significantly different at $p = 0.05$ based on least squares means differences. Where no lowercase letters follow the means, no significant differences were found.

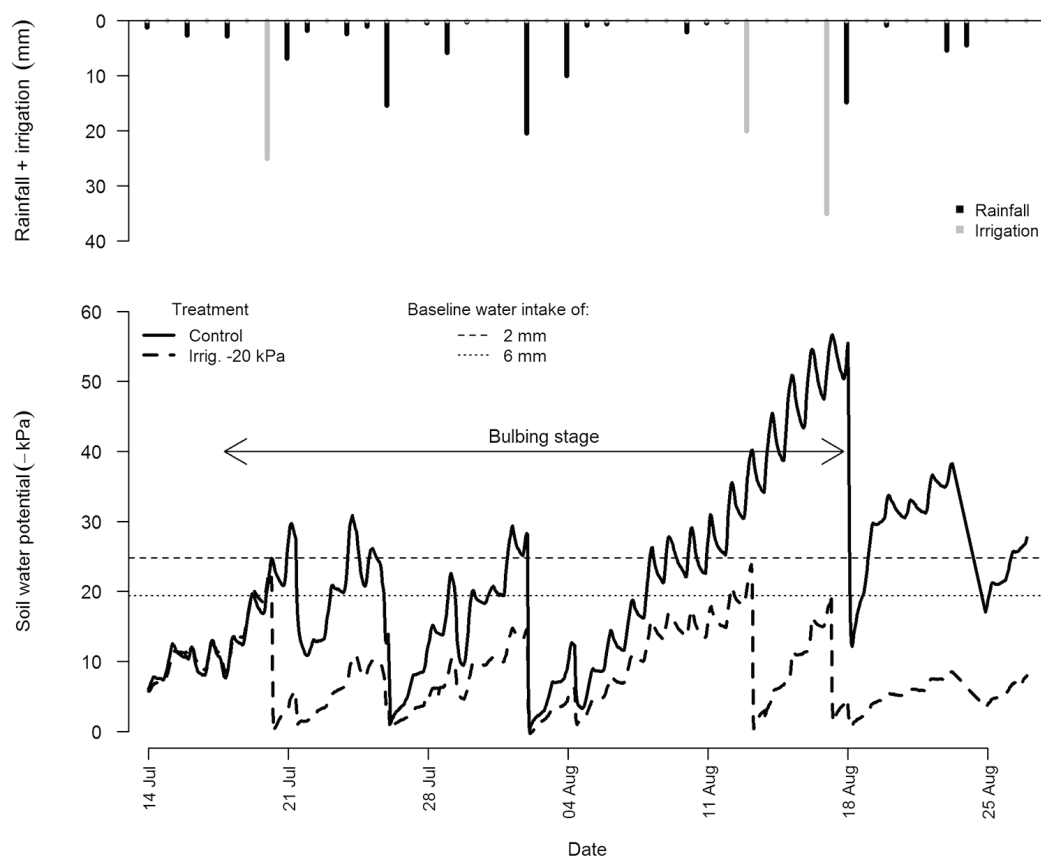


Fig. 4. Soil water potential at a 0.2-m depth with time, showing the bulbing stage critical period for yellow onion (cultivar Timberline) irrigated at -20 kPa in 2009.

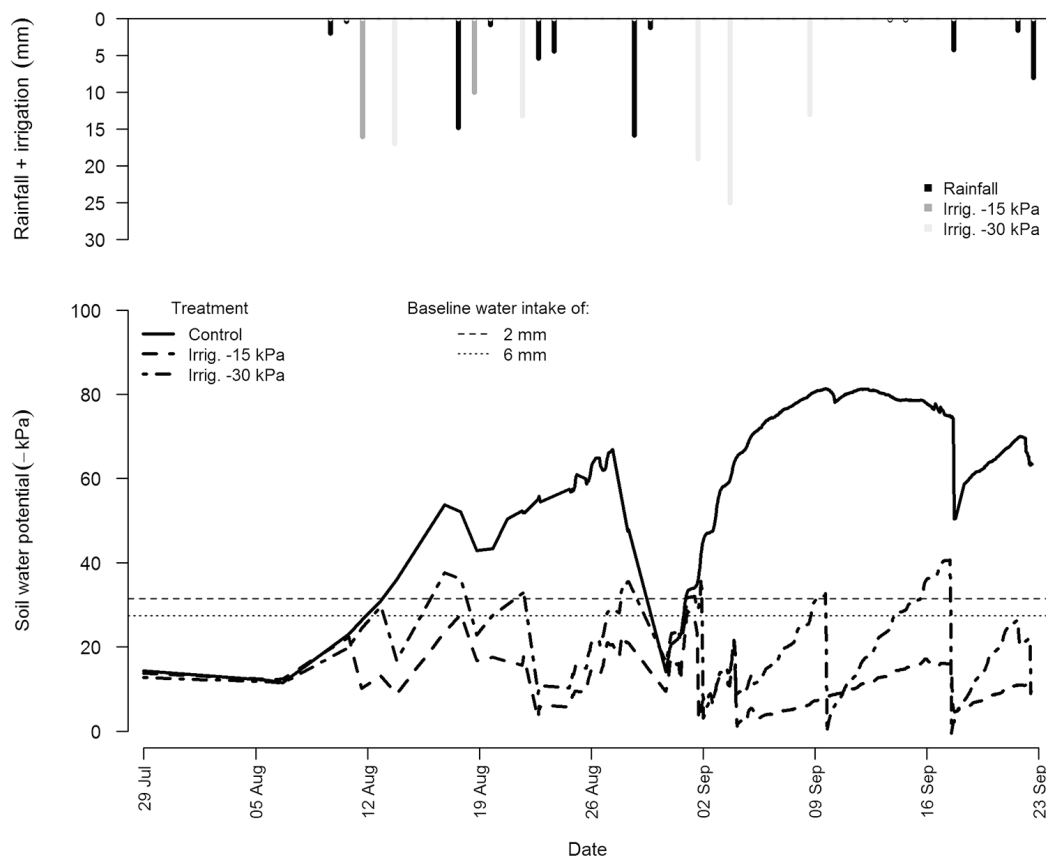


Fig. 5. Seasonal profile of soil water potential in the root zone of celery (cultivar XP266) irrigated at -15 and -30 kPa in 2009.

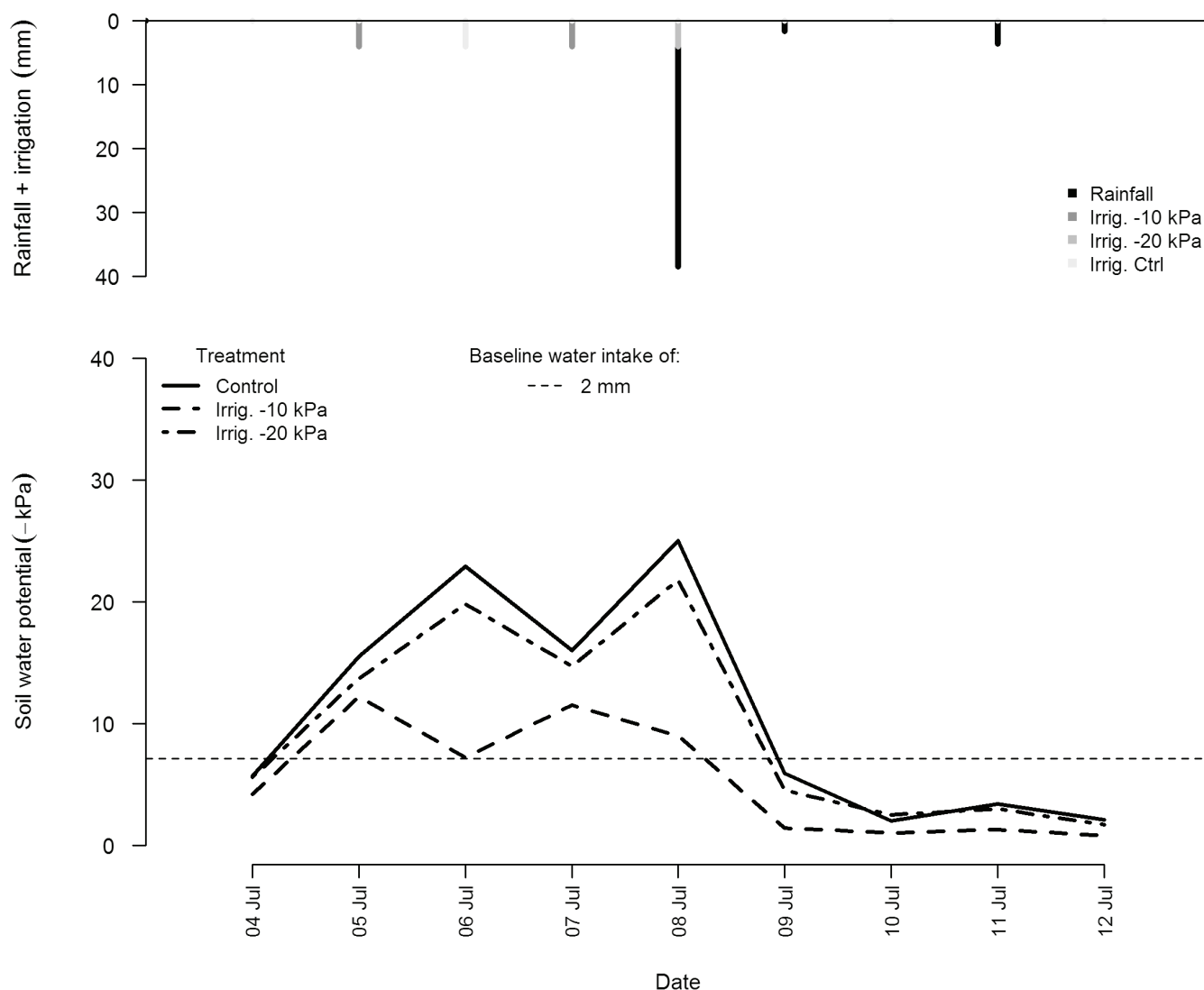


Fig. 6. Measured soil water potential at a 0.03-m depth in Field 4 for spinach (cultivars Waitiki, C2–606, and Toucon all pooled together) irrigated at –10 and –20 kPa in 2010.

Spinach was irrigated one to four times per week, and the treatment with the irrigation set point of –10 kPa received the highest amount of irrigation water (Table 4). The –20 kPa and control treatments had drier conditions, resulting in similar soil water potentials that averaged –19 and –16 kPa, respectively (Fig. 6), probably experiencing water stress, as outlined below. Three of the eight fields were not irrigated because an average of 30 mm of rain fell during the week following sowing and thus none of the thresholds was reached. Hence, depending on the year, irrigation resulted in different soil water potentials among treatments, with possible yield implications.

Plant Growth, Yield, and Water Use Efficiency

The irrigation treatments affected the yields of the different crops, depending on the year.

Onion

In both years, irrigation at a threshold of –20 kPa during the bulbing stage significantly increased total yield compared with the unirrigated treatment (Table 2). When onion yields were sorted by size class, irrigation level did not affect the small

and medium onion sizes in either year, although it did affect the prepack size in 2008. For the jumbo size class (interesting from a commercial point of view), the irrigated plots produced significantly higher yields in both years during the bulbing stage (6.3 and 7.1 Mg ha^{–1} in 2008 and 2009, respectively) compared with the unirrigated plots (2.1 and 1.9 Mg ha^{–1} in 2008 and 2009, respectively). This observation is an indication that larger onion sizes can be produced when more water is applied, which is in agreement with the results of Enciso et al. (2009), Kruse et al. (1987), and Shock et al. (1998, 2000a, 2000b), who obtained higher total marketable, jumbo, and colossal yields with wetter treatments. This finding is of commercial interest because jumbo onions have greater market value. Irrigation of onions during the bulbing stage would therefore be a great commercial strategy.

The lower yields of the unirrigated plants could be attributed to a water shortage during bulb formation and enlargement because onions at those stages are particularly sensitive to moisture deficits, which lead to a higher percentage of smaller sized bulbs (Martin de Santa Olalla et al., 2004). Similar observations were made by Pelter et al. (2004), who found that longer water stress periods from the seven-leaf treatment resulted

in small onion bulbs. In both years, WUE was the highest for the unirrigated treatment, but no significant differences were observed (Table 1).

It is interesting to note that even if the total amount of precipitation during the growing season was adequate, unpredictable rainfall events resulted in temporarily high soil water potentials that may have occurred during the critical bulbing stage of onion, reducing yields of jumbo onions. These results are in agreement with those of Shock et al. (1998), who emphasized that under warm and dry growing conditions, an irrigation threshold higher than -12.5 kPa was recommended. The real benefit in applying water in greater amounts than required for ET replacement must be compatible with the best storage performance, and the high prices of jumbo onions would compensate for more irrigation.

Celery

In 2008, no significant differences were observed among all treatments in terms of untrimmed and trimmed weight per plant, total plant height, plant weight, and number of marketable petioles (Table 3). This result was probably attributable to sufficient rainfall, higher than ETc, because the threshold of -50 kPa was never reached during this season, and as a result the small amount of water supplied by irrigation had no significant effect. In 2009, limited rainfall (78.9 mm below the 16-yr average) occurred during the vegetative growth phase and the last 4 wk before harvest, and the treatment thresholds were reached on multiple occasions (Fig. 5). Parameters usually contributing to high yields (untrimmed plant height and stalk diameter) were significantly higher in the irrigated than the unirrigated plots (Table 3). In general, the unirrigated plants were found to be significantly shorter and narrower, reflecting the moisture deficits in the unirrigated plots. The supplemental irrigation (at -15 or -30 kPa) significantly increased the weight of trimmed and untrimmed plants by 18 and 43%, respectively. The increased weight of the irrigated plants might be due to the observed growth of taller plants with more leaves and larger stalks (Breschini and Hartz, 2002), causing faster synthesis and translocation of photosynthesis products from source to sinks. A similar effect of irrigation on celery leaf yield was also observed by Rožek (2007) under a drip irrigation system with a total yield increase of 55.9% in comparison with unirrigated plants. The WUE ranged from 0.21 to 0.26 Mg ha⁻¹ mm⁻¹ in 2009, similar to the values obtained for celery by Todorovic et al. (2005). The WUE was higher for the control than the -15 kPa treatment (Table 1), indicating that the WUE increased because of the reduction in irrigation. Moreover, the highest celery yield was obtained when the water supply was increased because the marketable parts of celery have very high water content (about 85–95%).

Spinach

Spinach seedling emergence varied from 417 to 798 plants m⁻² for the -10 kPa treatment, 379 to 748 plants m⁻² for the -20 kPa treatment, and 367 to 716 plants m⁻² for the control treatment (Table 4). The low germination rate recorded in Field 4 was associated with waterlogging of the soil after heavy rain, when many seeds failed to germinate in comparison with the starting seed density (847 seeds m⁻²). Inhibition of the germination of

spinach seeds in the presence of excessive moisture has been attributed to the production of mucilage, which forms a barrier to diffusion of sufficient O₂ for germination (Heydecker and Orphanos, 1968). Only two of the five irrigated fields showed differences. In Fields 2 and 6, the wettest treatment (-10 kPa) had higher germination rates (650 and 798 plants m⁻², respectively), with an average increase of 9.1% compared with the combined mean of the other two treatments (Table 4). In contrast, in Fields 4, 7, and 8, irrigation did not affect spinach seed germination even though almost as much water was applied in wet treatments to those fields as in Fields 2 and 6. This difference may be partly explained by the fact that seeds germinated under conditions of decreasing temperatures in the spring vs. conditions of increasing temperatures in the summer. According to Finch-Savage and Phelps (1993), the mean rate and distribution of seedling emergence are determined largely by temperature and influenced second by soil water potential. The average germination response of all sites was statistically significant and tended to be slightly higher by 6% (661 plants m⁻²) when more water was used, as with the -10 kPa treatment compared with the other two treatments (Table 4). There were no differences in the germination rate between the -20 kPa and the control treatments because approximately similar soil water potential was observed (Fig. 6). Al-Rawi and Abdel (2011) reported similar findings for lentil (*Lens culinaris* Medik.), mung bean [*Vigna radiata* (L.) R. Wilczek var. *radiata*], and common vetch (*Vicia sativa* L. ssp. *sativa*), where a higher germination percentage was seen when the soil was moistened to 100% of its available water capacity compared with 25, 50, and 75%. In the present study, no significant difference in yield among treatments was found in each field, but the average yields of the two fields where yields could be measured showed significant increases of 11 and 7% at the threshold of -10 kPa relative to the threshold of -20 kPa and the control, respectively.

Water Desorption Curve and Unsaturated Hydraulic Conductivity

Because yields were shown to be affected by the irrigation threshold, it would require years of experiment to find the optimal threshold. Practically, this threshold can be predicted without having to run such experiments. Soil water retention curves and the associated unsaturated hydraulic conductivities for onion, celery, and spinach seed fields indicated volumetric water contents in the range 0.60 to 0.90 cm³ cm⁻³, data consistent with values reported for muck soils (Boelter and Blake, 1964) (Fig. 1–3). The unsaturated hydraulic conductivity (empirical and fitting parameters shown in Table 5) showed a decrease with decreasing matric potential and a piecewise exponential shape. In such a case, the exponential part of the shape was fitted (see appendix) into the -50 to -500 cm range (see inserts on Fig. 1 and 2) because it almost corresponded to the range in matric potential (-5 to -50 kPa) encountered and it allowed the use of Eq. [A5] (see appendix for that range).

These soil hydraulic properties were necessary to calculate critical thresholds at which a 2 or a 6 mm d⁻¹ ET would be met as in Eq. [A5] (Fig. 1 and 2) or to directly identify the matric potential threshold for a 2 mm d⁻¹ ET met for spinach seed germination (Fig. 3). The fact that a yield response was observed for onion in 2008 despite the fact that rainfall was higher than

Table 5. Empirical and fitting parameters obtained in HYDRUS for the Mualem–van Genuchten equation.

Crop and depth	Empirical parameters†			Fitting parameters‡			R ² predicted vs. observed
	θ_s cm ³ cm ⁻³	K_s cm d ⁻¹	θ_r cm ³ cm ⁻³	α cm ⁻¹	n	L cm	
Onion							
15–30 cm	0.79	9.3	0.05	0.0157	1.1524	0.5	0.9982
30–45 cm	0.82	21.9	0.05	0.0217	1.1087	0.5	0.9979
Celery							
15–30 cm	0.82	56.0	0.05	0.0213	1.1425	0.5	0.9996
30–45 cm	0.87	408.5	0.05	0.0565	1.1299	0.5	0.9991
Spinach seeds							
15–30 cm	0.82	26.4	0.05	0.0183	1.2048	0.5	0.9989

† θ_s , soil saturated water content; K_s , saturated hydraulic conductivity; θ_r , soil residual water content.

‡ α , fitting parameter in the Mualem–van Genuchten equation representing the inverse of the air-entry value; n , fitting parameter representing a pore-size distribution index; L , pore-connectivity parameter in the hydraulic conductivity function, usually estimated at 0.5 for many soils

ETc (Tables 1 and 2) and that, for celery in 2009, differences in terms of untrimmed plant height and stalk diameter (Table 3) were also observed between the irrigated and the unirrigated treatments even though the total amount of water applied for both irrigated treatments was higher than the ETc (Table 1) underlines the relevance of having an adequate supply of water at all critical times of crop growth (Breschini and Hartz, 2002; Pelter et al., 2004). It can be hypothesized that the proper amount of water may not have been applied and been available at those critical times in the control treatments. The calculated average ETc oscillated between 2.3 and 3.0 mm d⁻¹ for both crops and both years, consistent with the results of Gallichand et al. (1991). On the same soil type and when average water fluxes during capillary rise are calculated (Table 1), it can be seen that, on average and outside of rainfall events, when ETc was probably highest, the capillary rise differed among treatments and was considerably below the average seasonal ETc at different depths. This factor was investigated further with more accuracy using Eq. [A5], and comparisons of soil water potentials observed at all times in yield-responding years were performed for all three crops to calculate the critical threshold at which a 2 or a 6 mm d⁻¹ ET would be met throughout the whole root zone domain. Such data are reported in Fig. 4 and 5. It is easily observed that the top yield treatment was always within potential ranges higher than the horizontal line corresponding to the soil water potential with enough upward matric flux to meet a 2 or 6 mm d⁻¹ ET demand (Fig. 4 and 5). For spinach, the line corresponding to a near-surface flux of 2 mm d⁻¹ at the 3-cm depth was drawn on Fig. 6. It indicates that the drier treatment was sometimes above the 2 mm d⁻¹ flux threshold and the corresponding unsaturated hydraulic conductivity (Fig. 3), suggesting a lack of moisture to the surface. Figure 4 to 6 also suggest that stresses may be limited to a short period of time, leading to significant yield effects. These results are therefore consistent with the findings of Périard Larrivé (2012), who suggested, using hourly data, that bulk water fluxes from the soil are limiting to plant growth in that soil type (–10 to –15 kPa). These results are also consistent with findings for nursery plants (Caron et al., 1998) and *Petunia × hybrida* plants in coarse organic growing media (Jobin et al., 2004). In addition, the ranges of optimal thresholds are consistent with values reported for coarse mineral soils by Shock and Wang (2011) for the drip irrigation of onion, lettuce, and celery.

It may be somewhat surprising that such a simple, steady-state approach with a daily time step could help identify the onset of stress, given that complex hourly transient-state numerical models have been built to determine the thresholds for lettuce (Périard Larrivé, 2012). However, a similar daily-step, steady-state approach was successfully used for the prediction of water potential in a drying profile and signs of plant stress in coarse growing media made of a sphagnum, bark, and muck soil blend (Caron et al., 2005). Given that deriving thresholds requires very significant effort, it appears relevant to pursue additional research into establishing links between yield and soil hydraulic properties to derive critical irrigation thresholds that could be adjusted dynamically according to ET, root depth, and crop growth stage for other soil types.

Three main conclusions were reached:

1. Irrigation initiated at –20 kPa during bulb formation increased the yield of jumbo onions and water stress occurring during the critical bulbing stage of onion led mostly to small-sized bulbs. This study also recorded greater yield, plant height, and stalk diameter in celery with irrigation thresholds of –15 and –30 kPa relative to the control treatment and for spinach, the average germination of all fields tended to be slightly higher by 6% (661 plants m⁻²) with more water for the irrigation threshold of –10 kPa compared with the –20 kPa and control treatments.
2. Overall, even if the total amount of precipitation during the growing season was adequate, unpredictable time intervals between rainfall events may result in temporary soil moisture deficits, which may reduce yield and quality, particularly if the water stress occurs during the critical stage of a crop. There were no differences between the –15 to –30 kPa sprinkler-irrigated treatments compared with the unirrigated control, which had an average potential of –66 kPa.
3. The limiting threshold values were consistent with critical threshold estimates derived from ET estimates and soil hydraulic properties using a simple analytical solution to water flow equations. Hence, predicting irrigation thresholds using a simple analytical solution could be useful to any scientist or grower who is interested in irrigation of shallow root system crops grown on muck soils or even mineral soils by knowing a priori at least the soil hydraulic properties.

APPENDIX

Determination of Critical Soil Pressure Head in the Root Zone

Considering a root water uptake constant S_0 ($\text{cm}^3 \text{cm}^{-3} \text{d}^{-1}$) for the depth interval (cm) $0 \leq z \leq L$, $z = 0$ representing the bottom of the root zone and $z = L$ the soil surface (Fig. A1), the depth-dependant soil water flux potential $\Phi_s(z)$ ($\text{cm}^2 \text{d}^{-1}$) is given by (Yuan and Lu, 2005, Eq. [20]):

$$\Phi_s(z) = \frac{K_{sG} e^{\alpha^* (h_1 - z)}}{\alpha^*} + \frac{q_0 [e(-\alpha^* z) - 1]}{\alpha^*} + \frac{S_0 [(\alpha^* L + 1) e(-\alpha^* z) - \alpha^* (L - z) - 1]}{\alpha^{*2}} \quad [\text{A1}]$$

where K_{sG} (cm d^{-1}) is the fitted saturated hydraulic conductivity of the Gardner function, h_1 (cm) is the pressure head boundary condition at $z = 0$ cm, q_0 (cm d^{-1}) is the steady-state soil surface evaporation (evapotranspiration $[\text{ET}] = q_0 + S$, where $S = S_0 L$), and α^* (cm^{-1}) is the fitted exponent of the $K(h)$ Gardner function, described by

$$K(h) = K_{sG} e^{\alpha^* h} \quad [\text{A2}]$$

where h (–cm) is the pressure head under unsaturated conditions. The soil water flux potential $\Phi_s(z)$ at depth $z = 0$ is defined as

$$\Phi_s(z) = \frac{K_{sG} e^{\alpha^* h}}{\alpha^*} \quad [\text{A3}]$$

Substituting Eq. [A3] into Eq. [A1], we have

$$\frac{K_{sG} e^{\alpha^* h}}{\alpha^*} = \frac{K_{sG} e^{\alpha^* (h_1 - z)}}{\alpha^*} + \frac{q_0 (e^{-\alpha^* z} - 1)}{\alpha^*} + \frac{S_0 [(\alpha^* L + 1) e^{-\alpha^* z} - \alpha^* (L - z) - 1]}{\alpha^{*2}} \quad [\text{A4}]$$

Isolating h from Eq. [A4] obtains

$$h = \frac{1}{\alpha^*} \ln \left\{ \frac{1}{\alpha^* K_{sG}} \left[\frac{K_{sG} e^{-\alpha^* (-h_1 + z)} \alpha^* + q_0 \alpha^* e^{-\alpha^* z}}{-q_0 \alpha^* + S_0 e^{-\alpha^* z} \alpha^* L + S_0 e^{-\alpha^* z}} \right] \right\} \quad [\text{A5}]$$

where the argument of the natural logarithm should be positive. An analytical solution for this asymptote may then be derived by fixing the Real ($\mathbb{R}$ domain) limit of this argument to zero and solve it for h_1 :

$$h_1 = \frac{1}{\alpha^*} \ln \left[-\frac{1}{\alpha^* K_{sG}} \left(\frac{q_0 \alpha^* e^{-\alpha^* z} - q_0 \alpha^*}{+S_0 e^{-\alpha^* z} \alpha^* L + S_0 e^{-\alpha^* z} - S_0 \alpha^* L + S_0 \alpha^* z - S_0} \right) \right] \quad [\text{A6}]$$

and substituting $z = L$ in order to cover all the domain:

$$h_{1, \text{asymptotic}} = \frac{1}{\alpha^*} \ln \left[-\frac{1}{\alpha^* K_{sG}} \left(\frac{q_0 \alpha^* e^{-\alpha^* L} - q_0 \alpha^*}{+S_0 e^{-\alpha^* L} \alpha^* L + S_0 e^{-\alpha^* L} - S_0 \alpha^* L + S_0 \alpha^* z - S_0} \right) \right] + L \quad [\text{A7}]$$

which gives the asymptotic soil pressure head to maintain throughout all the domain as a function of soil properties and a constant root water uptake throughout the domain.

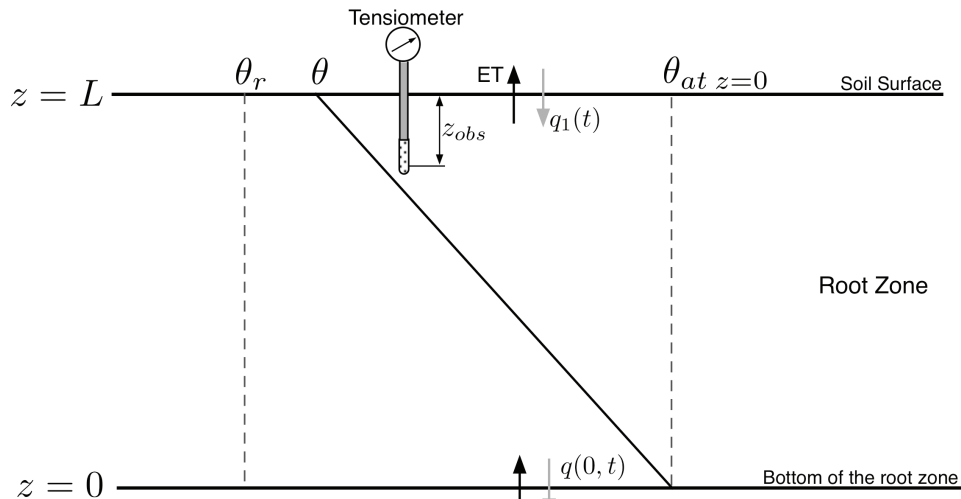


Fig. A1. Schematic view of a root zone domain with boundary conditions (adapted from Yuan and Lu, 2005), where z is the position in the soil profile, L is the domain depth (cm), θ is the volumetric soil water content ($\text{cm}^3 \text{cm}^{-3}$), θ_r is the volumetric residual soil water content (cm cm^{-3}), ET is the evapotranspiration rate (cm d^{-1}), and q is the water flux (cm d^{-1}).

In fact, the real domain of Eq. [A5] indicates the critical soil pressure head at the upper boundary L at which the soil is able to maintain a constant water uptake rate of S_0 . However, the critical soil pressure head at any measurement point z within the domain length $0 \leq z \leq L$ may be of interest. For that, Eq. [A5] needs to be used with $h_1 = h_{1,\text{asymptotic}}$. For example, considering the case of onion at 2 mm d^{-1} of evapotranspiration (Fig. 1), at $z = L$, with $L = 35 \text{ cm}$, $K_{sG} = 2.22 \times 10^{-1} \text{ cm d}^{-1}$, $\alpha^* = 0.01 \text{ cm}^{-1}$, we fixed q_0 at 0.1 cm d^{-1} , so then $S_0 = 0.002857 \text{ d}^{-1}$.

The first step is solving Eq. [A7] with these conditions:

$$h_{1,\text{asymptotic}} = \frac{1}{\alpha^*} \ln \left[-\frac{1}{\alpha^* K_{sG}} \left(\frac{q_0 \alpha^* e^{-\alpha^* L} - q_0 \alpha^*}{+S_0 e^{-\alpha^* L} \alpha^* L + S_0 e^{-\alpha^* L} - S_0} \right) \right] + L \quad [\text{A8}]$$

$$h_{1,\text{asymptotic}} = -128.14 \text{ cm}$$

And the second step is substituting $h_{1,\text{asymptotic}}$ (found in Eq. [A8]) into Eq. [A5] and solving at $z = z_{\text{obs}} = 20 \text{ cm}$:

$$h_c = \frac{1}{\alpha^*} \ln \left\{ \frac{1}{\alpha^* K_{sG}} \left[\frac{K_{sG} e^{-\alpha^* (-h_1 + z)} \alpha^* + q_0 \alpha^* e^{-\alpha^* z}}{-q_0 \alpha^* + S_0 e^{-\alpha^* z} \alpha^* L + S_0 e^{-\alpha^* z}} \right] \right\} \quad [\text{A9}]$$

$$h_c = -242.90 \text{ cm}$$

Every calculated critical pressure (matric) head (h_c) was then converted to potential units using $9.81 \text{ cm} = 1 \text{ kPa}$ in order to show critical threshold potentials (Fig. 1–6) to control irrigation using tensiometers.

Additionally, it can be shown that for steady-state conditions, the flux of water [$J_w(h)$, cm d^{-1}]:

$$J_w(h) = \frac{-d\Phi_s(h)}{dz} - \alpha^* \Phi_s(h) \quad [\text{A10}]$$

will tend toward $-\alpha^* \Phi_s(h)$, which can be shown to be equal to $-K(h)$ as the rooting depth approaches a plane. Hence, for a seed at a given depth, $K(h)$ can be a reasonable determination of the threshold with an ETc corresponding to an evaporation rate (e.g., 2 mm d^{-1}) and to the water flux to the surface, $J_w(h)$.

ACKNOWLEDGMENTS

We would like to thank many undergraduate students for their help in the field, as well as Yann Périard for his great collaborative work. We also acknowledge the financial support of the Natural Sciences and Engineering Research Council of Canada (NSERC) in the form of a Collaborative Research and Development Grant in partnership with Production Horticole Van Winden Inc, Les Fermes Hotte and Van Winden Inc, Maraichers J.P.L. Guérin et Fils Inc, Delfland Inc, and Vert Nature Inc.

REFERENCES

- Allen, R.G., L.S. Pereira, D. Raes, and M. Smith. 1998. Crop evapotranspiration: Guidelines for computing crop water requirements. Irrig. Drain. Pap. 56. FAO, Rome.
- Aloni, B., and E. Pressman. 1979. Petiole pithiness in celery leaves: Induction by environmental stresses and the involvement of abscisic acid. *Physiol. Plant.* 47:61–65. doi:10.1111/j.1399-3054.1979.tb06513.x
- Al-Rawi, I.M.T., and C.G. Abdel. 2011. Germination performance of some legume crops under varying soil water available capacities. *Int. J. Biosci.* 1:26–35.
- Boelter, D.H., and G.R. Blake. 1964. Importance of volumetric expressions of water contents of organic soils. *Soil Sci. Soc. Am. Proc.* 28:176–178. doi:10.2136/sssaj1964.03615995002800020015x
- Breschini, S.J., and T.K. Hartz. 2002. Drip irrigation management affects celery yield and quality. *HortScience* 37:894–897.
- Canadian Food Inspection Agency. 1999. Celery. CFIA, Ottawa, ON, Canada. www.inspection.gc.ca/food/fresh-fruits-and-vegetables/quality-inspection/vegetable-inspection-manuals/celery/eng/1303756803599/1303756936914 (accessed 18 Apr. 2013).
- Carminati, A., A.B. Moradi, D. Vetterlein, P. Vontobel, E. Lehmann, U. Weller, et al. 2010. Dynamics of soil water content in the rhizosphere. *Plant Soil* 332:163–176. doi:10.1007/s11104-010-0283-8
- Caron, J., D.E. Elrick, R. Beeson, and J. Boudreau. 2005. Defining critical capillary rise properties for growing media in nurseries. *Soil Sci. Soc. Am. J.* 69:794–806. doi:10.2136/sssaj2004.0108
- Caron, J., H.L. Xu, P.Y. Bernier, I. Duchesne, and P. Tardif. 1998. Water availability in three artificial substrates during *Prunus × cistena* growth: Variable threshold values. *J. Am. Soc. Hortic. Sci.* 123:931–936.
- Conseil des Productions Végétales du Québec. 1996. Grilles de référence en fertilisation. 2nd ed. AGDEX 540. Conseil des Productions Végétales du Québec, Québec City, QC, Canada.
- Dane, J.H., and J.W. Hopmans. 2002. Pressure cell. In: J.H. Dane and G.C. Topp, editors, *Methods of soil analysis. Part 4. SSSA Book Ser. 5. SSSA, Madison, WI*, p. 684–688.
- Elrick, D.E., and W.D. Reynolds. 1992. Methods for analyzing constant-head well permeameter data. *Soil Sci. Soc. Am. J.* 56:320–323. doi:10.2136/sssaj1992.03615995005600010052x
- Enciso, J., J. Jifon, and B. Wiedenfeld. 2007. Subsurface drip irrigation of onions: Effects of drip tape emitter spacing on yield and quality. *Agric. Water Manage.* 92:126–130. doi:10.1016/j.agwat.2007.05.017
- Enciso, J., B. Wiedenfeld, J. Jifon, and S. Nelson. 2009. Onion yield and quality response to two irrigation scheduling strategies. *Sci. Hortic.* 120:301–305. doi:10.1016/j.scienta.2008.11.004
- Fares, A., and A.K. Alva. 2000. Evaluation of capacitance probes for optimal irrigation of citrus through soil moisture monitoring in an Entisol profile. *Irrig. Sci.* 19:57–64. doi:10.1007/s002710050001
- Feigin, A., J. Letey, and W.M. Jarrell. 1982. Celery response to type, amount, and method of N-fertilizer application under drip irrigation. *Agron. J.* 74:971–977. doi:10.2134/agronj1982.00021962007400060011x
- Finch-Savage, W.E., and K. Phelps. 1993. Onion (*Allium cepa* L.) seedling emergence patterns can be explained by the influence of soil temperature and water potential on seed germination. *J. Exp. Bot.* 44:407–414. doi:10.1093/jxb/44.2.407
- Gallardo, M., L.E. Jackson, K. Schulbach, R.L. Snyder, R.B. Thompson, and L.J. Wyland. 1996. Production and water use in lettuces under variable water supply. *Irrig. Sci.* 16:125–137. doi:10.1007/BF02215620

- Gallichand, J., R.S. Broughton, J. Boisvert, and P. Rochette. 1991. Simulation of irrigation requirements for major crops in south western Québec. *Can. Agric. Eng.* 33:1–9.
- Gardner, W.R. 1958. Some steady-state solutions of the unsaturated moisture flow equation with application to evaporation from a water table. *Soil Sci.* 85:228–232. doi:10.1097/00010694-195804000-00006
- Hegde, D.M. 1986. Effect of irrigation regimes on dry matter production, yield, nutrient uptake, and water use of onion. *Indian J. Agron.* 31:343–348.
- Heydecker, W., and P.I. Orphanos. 1968. The effect of excess moisture on the germination of *Spinacia oleracea* L. *Planta* 83:237–247. doi:10.1007/BF00385333
- Hopmans, J.W., J. Šimůnek, N. Romano, and W. Durner. 2002. Inverse methods. In: J.H. Dane and G.C. Topp, editors, *Methods of soil analysis*. Part 4. SSSA Book Ser. 5. SSSA, Madison, WI. p. 963–1008.
- Jobin, P., J. Caron, P.-Y. Bernier, and B. Dansereau. 2004. Impact of two hydrophilic acrylic-based polymers on the physical properties of three substrates and the growth of *Petunia × hybrida* 'Brilliant Pink'. *J. Am. Soc. Hortic. Sci.* 129:449–457.
- Katzman, L.S., A.G. Taylor, and R.W. Langhans. 2001. Seed enhancements to improve spinach germination. *HortScience* 36:979–981.
- Kemble, J.K., and D.C. Sanders. 2000. Basics of vegetable crop irrigation. *Bull. ANR-1169*. Alabama Coop. Ext. Serv., Auburn. www.aces.edu/pubs/docs/A/ANR-1169/.
- Kruse, E.G., J.E. Ells, and A.E. McSay. 1987. Comparison of two onion irrigation scheduling programs. *J. Am. Soc. Hortic. Sci.* 112:738–742.
- Lascano, R.J., and C.H.M. van Bavel. 1984. Root water uptake and soil water distribution: Test of an availability concept. *Soil Sci. Soc. Am. J.* 48:233–237. doi:10.2136/sssaj1984.03615995004800020002x
- Leskovar, D.I., V. Esensee, and H. Belefant-Miller. 1999. Pericarp, leachate, and carbohydrate involvement in thermoinhibition of germinating spinach seeds. *J. Am. Soc. Hortic. Sci.* 124:301–306.
- Marouelli, W.A., and W.L.C. Silva. 2007. Water tension thresholds for processing tomatoes under drip irrigation in central Brazil. *Irrig. Sci.* 25:411–418. doi:10.1007/s00271-006-0056-6
- Martin de Santa Olalla, F., A. Dominguez-Padilla, and R. Lopez. 2004. Production and quality of onion crop (*Allium cepa* L.) cultivated under controlled deficit irrigation conditions in a semi-arid climate. *Agric. Water Manage.* 68:77–89. doi:10.1016/j.agwat.2004.02.011
- Pelter, G.Q., R. Mittelstadt, B.G. Leib, and C.A. Redulla. 2004. Effects of water stress at specific growth stages on onion bulb yield and quality. *Agric. Water Manage.* 68:107–115. doi:10.1016/j.agwat.2004.03.010
- Périard, Y., J. Caron, S. Jutras, J.A. Lafond, and A. Houliot. 2013. Irrigation management of romaine lettuce in Histosols at two spatial scales: Water, energy, leaching and yield impacts. In: H. Bjornlund et al, editors, *Sustainable Irrigation and Drainage IV: Management, Technologies and Policies*, Adelaide, SA, Australia. 11–13 Dec. 2012. WIT Press, Southampton, UK. p. 171–188.
- Périard Larrivée, Y. 2012. Irrigation spatio-temporelle de la laitue romaine en sol organique. M.S. thesis. Laval Univ., Quebec City, QC, Canada.
- Phene, C.J., and O.W. Beale. 1976. High-frequency irrigation for water nutrient management in humid regions. *Soil Sci. Soc. Am. J.* 40:430–436. doi:10.2136/sssaj1976.03615995004000030034x
- Phene, C.J., and T.A. Howell. 1984. Soil sensor control of high frequency irrigation systems. *Trans. ASAE* 27:392–396. doi:10.13031/2013.32798
- Reynolds, W.D., and D.E. Elrick. 2002. Constant head soil core (tank) method. In: J.H. Dane and G.C. Topp, editors, *Methods of soil analysis*. Part 4. SSSA Book Ser. 5. SSSA, Madison, WI. p. 804–808.
- Rožek, E. 2007. Reaction of leaf celery (*Apium graveolens* L. var. *secalinum*) to planting density and irrigation. *Veg. Crops Res. Bull.* 66:69–77.
- SAS Institute. 2008. The SAS system for Windows. Release 9.2. SAS Inst., Cary, NC.
- Saxton, A.M. 1998. A macro for converting mean separation output to letter groupings in PROC MIXED. In: *Proceedings of the 23rd Annual SAS Users Group International*, Nashville, TN. 22–25 Mar. 1998. SAS Inst., Cary, NC. p. 1243–1246.
- Shock, C.C., E.B.G. Feibert, and L.D. Saunders. 1998. Onion yield and quality affected by soil water potential as irrigation threshold. *HortScience* 33:1188–1191.
- Shock, C.C., E.B.G. Feibert, and L.D. Saunders. 2000a. Irrigation criteria for drip-irrigated onions. *HortScience* 35:63–66.
- Shock, C.C., E.B.G. Feibert, and L.D. Saunders. 2000b. Onion storage decomposition unaffected by late-season irrigation reduction. *HortTechnology* 10:176–178.
- Shock, C.C., A.B. Pereira, B.R. Hanson, and M.D. Cahn. 2007. Vegetable irrigation. In: R.J. Lascano and R.E. Sojka, editors, *Irrigation of agricultural crops*. 2nd ed. Agron. Monogr. 30. ASA, CSSA, and SSSA, Madison, WI. p. 535–606.
- Shock, C.C., and F.-X. Wang. 2011. Soil water tension, a powerful measurement for productivity and stewardship. *HortScience* 46:178–185.
- Soil Classification Working Group. 1998. The Canadian system of soil classification. 3rd ed. Publ. 1646. NRC Res. Press, Ottawa, ON, Canada.
- Soil Survey Staff. 1999. Soil taxonomy: A basic system of soil classification for making and interpreting soil surveys. 2nd ed. Agric. Handbk. 436. U.S. Gov. Print. Office, Washington, DC.
- Todorovic, M., A. Caliendo, and R. Albrizio. 2005. Irrigated agriculture and water use efficiency in Italy. *Options Méditerranéennes* 57:102–138.
- Yuan, F., and Z. Lu. 2005. Analytical solutions for vertical flow in unsaturated, rooted soils with variable surface fluxes. *Vadose Zone J.* 4:1210–1218. doi:10.2136/vzj2005.0043